

Logistic Regression

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Introduction to Binary Outcomes

Continuous vs. Categorical Variables

- General linear regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- Independent variables (x 's):
 - Continuous: age, income, height → use numerical value.
 - Categorical: gender, city, ethnicity → use dummies.
- Dependent variable (y):
 - Continuous: consumption, time spent → use numerical value.
 - Categorical: yes/no → use dummies.

Examples of Binary Outcomes

- Should a bank give a person a loan or not?
- Is an individual transaction fraudulent or not?
- What determines admittance into a school?
- Which people are more likely to vote against a new law?
- Which customers are more likely to buy a new product?

Representing the Binary Outcomes

- There are two outcomes: Yes and No
- We will create a dummy variable to indicate if an observation is a Yes or a No:
 - $y = 1$ if Yes
 - $y = 0$ if No
- If we code the variable the other way around, our coefficients will have the same magnitudes but opposite signs.

A linear model?

- Aside from being binary, there's really nothing special about our dependent variable (y).
- Its value is higher (from a 0 to a 1) if a customer subscribes, so whatever makes it higher increases the likelihood of subscription.
- We can then run:

$$\textit{subscribe} = \beta_0 + \beta_1 \textit{age} + \varepsilon$$

Result of Linear Model

gretl: model 1				
File Edit Tests Save Graphs Analysis LaTeX				
Model 1: OLS, using observations 1-1000				
Dependent variable: subscribe				
	coefficient	std. error	t-ratio	p-value
-----	-----	-----	-----	-----
const	-1.70073	0.0638035	-26.66	1.20e-118 ***
age	0.0645433	0.00178736	36.11	2.52e-183 ***
Mean dependent var	0.573000	S.D. dependent var	0.494890	
Sum squared resid	106.0736	S.E. of regression	0.326016	
R-squared	0.566464	Adjusted R-squared	0.566030	
F(1, 998)	1304.002	P-value(F)	2.5e-183	
Log-likelihood	-297.1275	Akaike criterion	598.2550	
Schwarz criterion	608.0705	Hannan-Quinn	601.9855	

$$\text{subscribe} = -1.700 + 0.064 \text{ age}$$

Interpreting the Result

- If our dependent variable is binary, then we want to see what makes it change from a 0 to 1.
- This can also be interpreted as what increases the likelihood of subscription, or $P(\textit{subscribe} = 1)$, which we can also simply denote as p .
- The result can be read as:
$$P(\textit{subscribe} = 1) = p = -1.700 + 0.064 \textit{age}$$
- Every additional year of $\textit{age}$ increases the probability of subscription by 6.4%.

Problems with the Linear Approach

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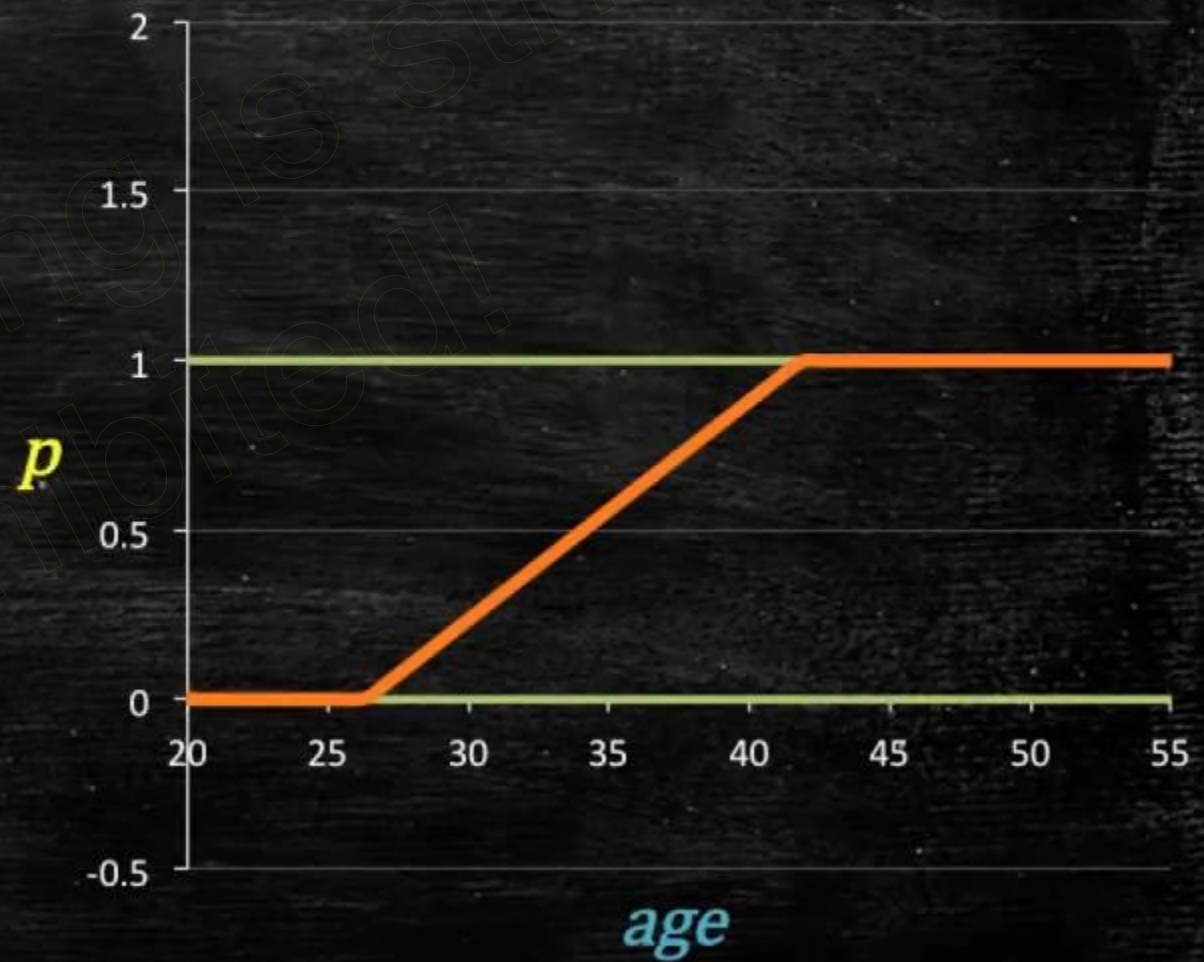
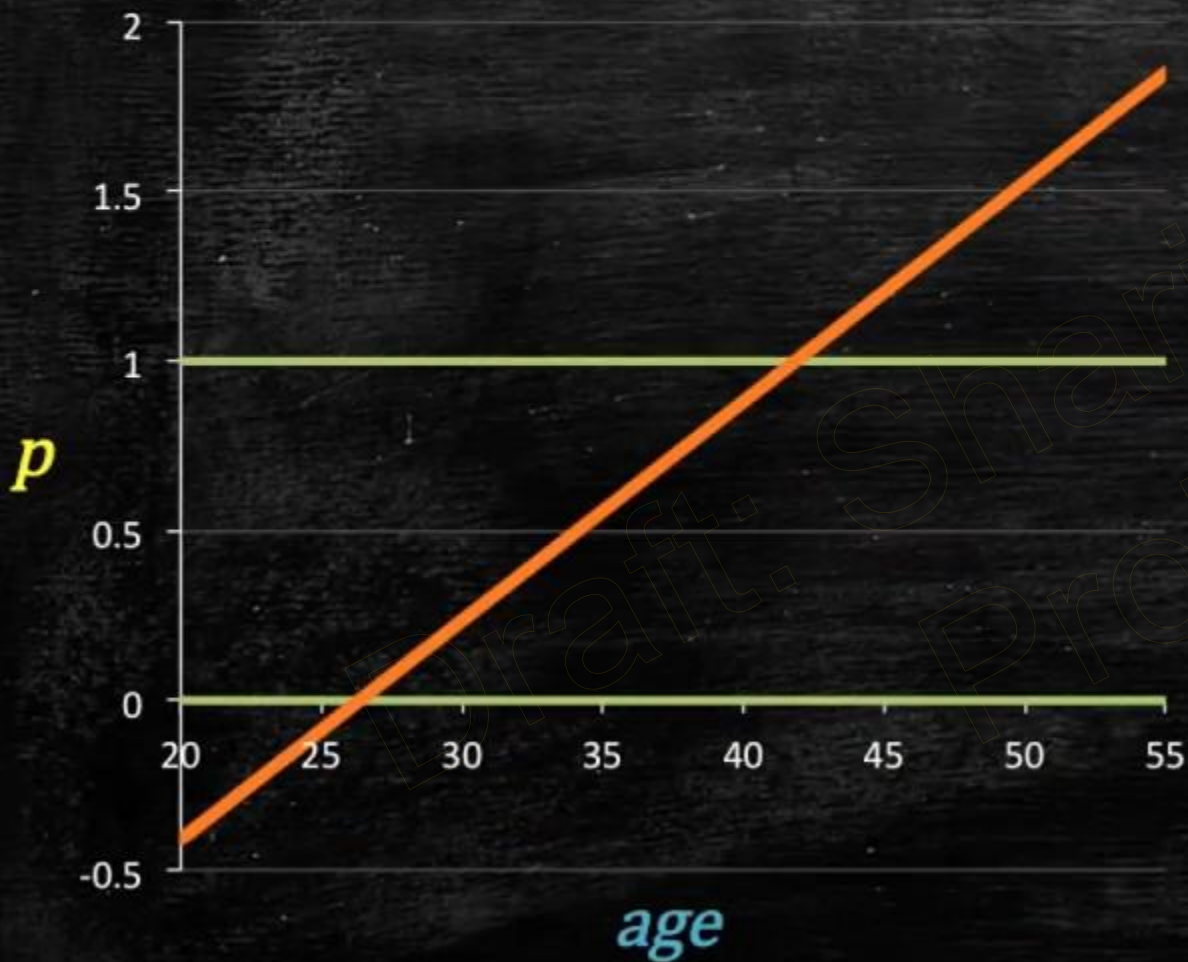
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- What about people with 25 and 45 years of age?

$$p = -1.700 + 0.064 \times 25 = -0.09$$

$$p = -1.700 + 0.064 \times 45 = 1.20$$

Linear Model Plot

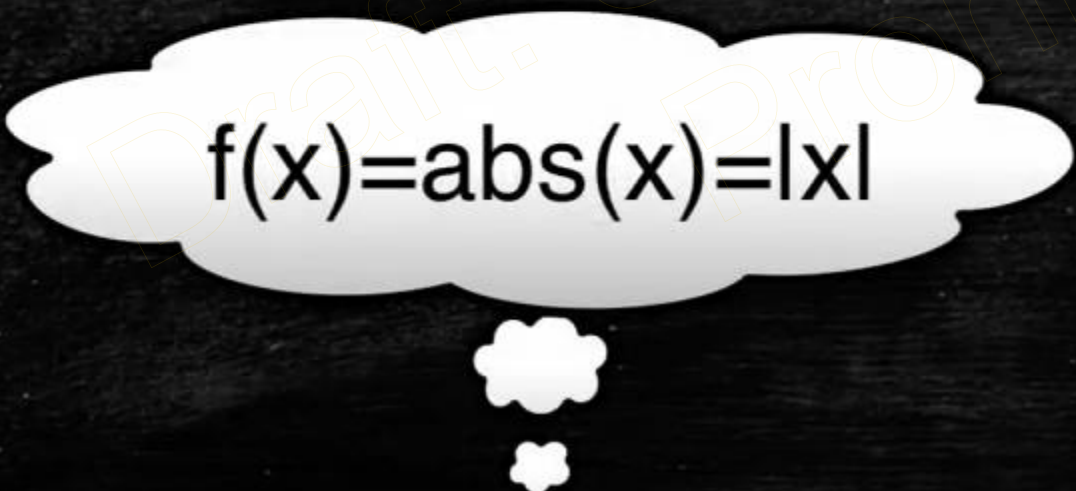


Two Steps!

1. It must always be positive (since $p \geq 0$)

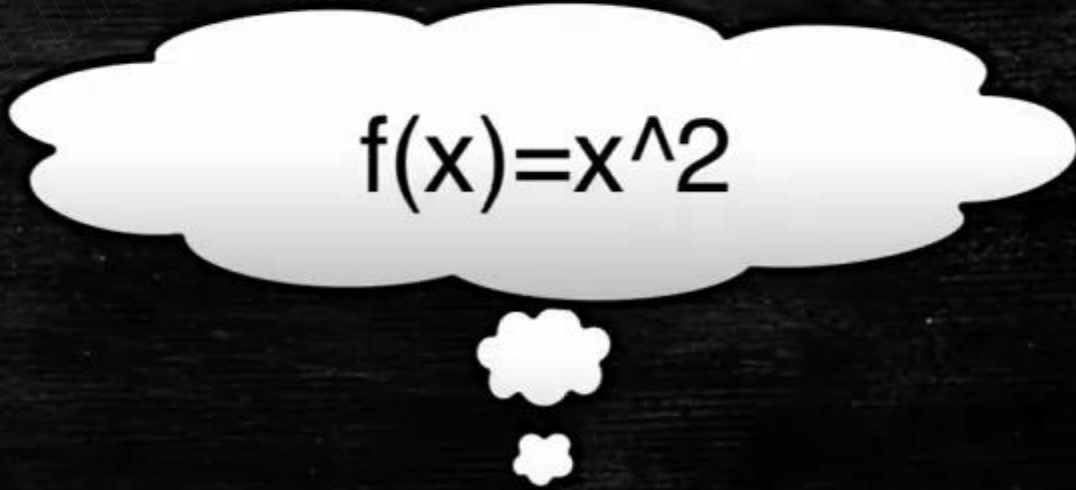
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$$f(x) = \text{abs}(x) = |x|$$

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$$f(x)=x^2$$

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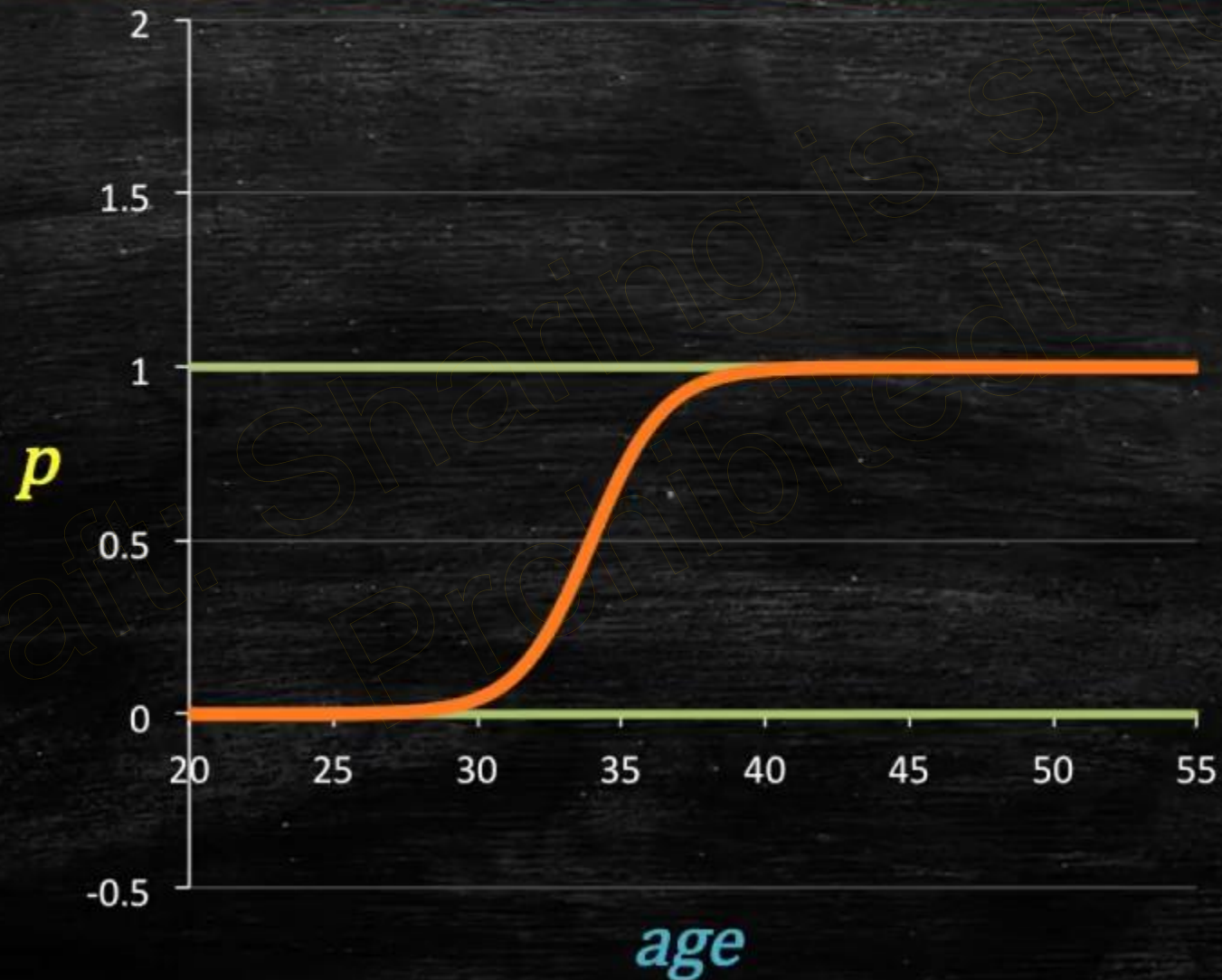
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$$p = \frac{\exp(\beta_0 + \beta_1 \text{age})}{\exp(\beta_0 + \beta_1 \text{age}) + 1} = \frac{e^{\beta_0 + \beta_1 \text{age}}}{e^{\beta_0 + \beta_1 \text{age}} + 1}$$

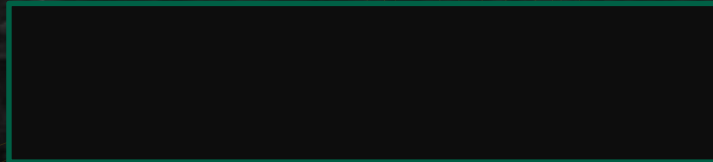
Logistic Model Plot



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Log Odds

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- Even though the probability of a customer subscribing (p) is not a linear function of age, the simple transformation is a linear function of age.

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- Even though the probability of a customer subscribing (p) is not a linear function of age, the simple transformation is a linear function of age.
- The above equation is the one used in **logistic regressions**.

The Estimated Logistic Model

- The estimated model was:

$$\ln\left(\frac{p}{1-p}\right) = -26.52 + 0.78 \text{ age}$$

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$$p = \frac{\exp(-26.52 + 0.78 \text{ age})}{\exp(-26.52 + 0.78 \text{ age}) + 1} = \frac{e^{-26.52 + 0.78 \text{ age}}}{e^{-26.52 + 0.78 \text{ age}} + 1}$$

Logistic Regression

Interpretation of Coefficients and Forecasting

Leveraging the Similarities with Linear Models

gretl: model 2

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Model 2: Logit, using observations 1-1000
Dependent variable: subscribe
Standard errors based on Hessian

	coefficient	std. error	z	slope
const	-26.5240	1.82819	-14.51	
age	0.781053	0.0535623	14.58	0.154207

Mean dependent var	0.573000	S.D. dependent var	0.494890
McFadden R-squared	0.636613	Adjusted R-squared	0.633683
Log-likelihood	-247.9937	Akaike criterion	499.9873
Schwarz criterion	509.8028	Hannan-Quinn	503.7179

Number of cases 'correctly predicted' = 884 (88.4%)
 $f(\text{beta}'x)$ at mean of independent vars = 0.197
Likelihood ratio test: Chi-square(1) = 868.915 [0.0000]

	Predicted	
	0	1
Actual 0	350	77
1	39	534

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Sign of coefficients still represents a positive or negative influence on dependent variable.

Leveraging the Similarities with Linear Models

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Standard errors can be used to estimate confidence intervals:

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$$0.78105 \pm 2 \times 0.05356$$
$$[0.674, 0.888]$$

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$$\ln\left(\frac{p}{1-p}\right) = -26.524 + 0.781 \text{ age}$$

- For every unit increase of *age*, $\ln\left(\frac{p}{1-p}\right)$ increases 0.78 units.

Increasing $\ln(\text{odd})$ is actually increasing probability.

THANK YOU!